

Annotation Validation Report

Comparing Automated Topic Analysis Against Human Annotation

This report evaluates two automated topic classification methods — a keyword-based pipeline and a semantic (AI embedding) tagger — against human annotation across 400 labeled articles and 12 topics.

Annotator 1 and Annotator 3 each independently labeled a separate set of 200 articles from the Black media corpus.

400

Articles Evaluated

12

Topics

2

Human Annotators

2

Pipeline Methods

What We Are Comparing

Two automated pipelines were validated against hand-labeled articles.

Method 1 — Keyword Pipeline

Scans each article's title, description, and body text for a curated dictionary of seed phrases (e.g. 'police brutality', 'voter suppression', 'redlining'). If a phrase is found, the article is tagged with that topic.

Strengths: Fast, transparent, easy to audit.

Weaknesses: Misses articles that use different wording for the same concept.

Method 2 — Semantic Tagger

Uses an AI sentence-embedding model (all-MiniLM-L6-v2) to measure the meaning similarity between each article and each topic description. Tags the article if similarity exceeds a threshold.

Strengths: Captures meaning, not just exact words.

Weaknesses: Less transparent; threshold-sensitive.

Human Annotation Setup

Two trained annotators independently labeled separate sets of 200 articles each (400 total). For each article they marked which of the 12 social conflict topics were present using a structured Excel sheet.

The human labels serve as the 'ground truth' against which both automated pipelines are evaluated. Because the annotators labeled different articles, results are reported separately for each annotator's set.

What Are We Measuring?

The core question: how closely does the computer agree with a trained human reader?

STEP 1 Human Reads & Labels

Two trained annotators (Annotator 1 and Annotator 3) each manually read 200 articles from Black media publications.

For every article they answered:
Which of the 12 social conflict topics is this article about?

They checked each topic that applied.
These human labels are our GROUND TRUTH — the benchmark we hold the computer against.

STEP 2 Computer Labels the Same Articles

We ran the exact same 400 articles through both automated pipelines:

- Keyword Pipeline — looks for seed phrases
- Semantic Tagger — uses AI embeddings

Each pipeline independently decides:
which of the 12 topics does this article belong to?

The computer has NOT seen the human labels.

STEP 3 Compare & Score

For every article and every topic we ask:
Did the computer agree with the human?

- ✓ Both said YES → True Positive
- ✓ Both said NO → True Negative
- ✗ Computer YES, Human NO → False Positive
- ✗ Computer NO, Human YES → False Negative

Precision, Recall, F1, and Kappa are all calculated from these four counts.

Important: the two annotators labeled completely different sets of articles.

Results are therefore reported separately for each annotator's 200-article set. This is not a flaw — it shows how consistent the pipeline is across different article samples.

Understanding Cohen's Kappa

Why Kappa — and not just accuracy — is the right metric for this study.

The Problem With Simple Accuracy

Imagine the pipeline NEVER tags any article with "Reparations".
Out of 200 articles, only 3 were labeled Reparations by the human.

Accuracy = 197 correct out of 200 = 98.5% ✓

But the pipeline learned nothing — it just got lucky
because the topic is rare. Accuracy is misleading
whenever one label (NO) is much more common than the other (YES).

In our 12-topic study, most articles are NOT about any given topic,
so a pipeline that tags nothing would still look ~90% accurate.

What Kappa Does Instead

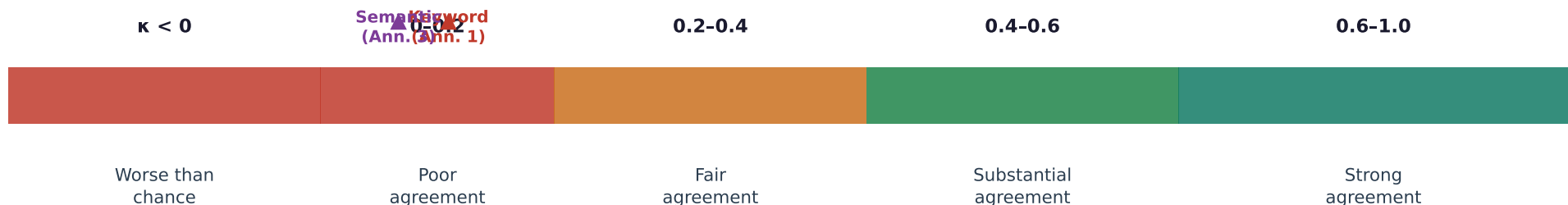
Kappa asks: how much better than chance is the agreement?

It first calculates how often we'd expect the computer
and human to agree purely by random chance,
based on how often each label appears.

Then it measures how much the ACTUAL agreement
exceeds that chance baseline.

Formula:

$$\kappa = (\text{Observed Agreement} - \text{Chance Agreement}) \div (1 - \text{Chance Agreement})$$

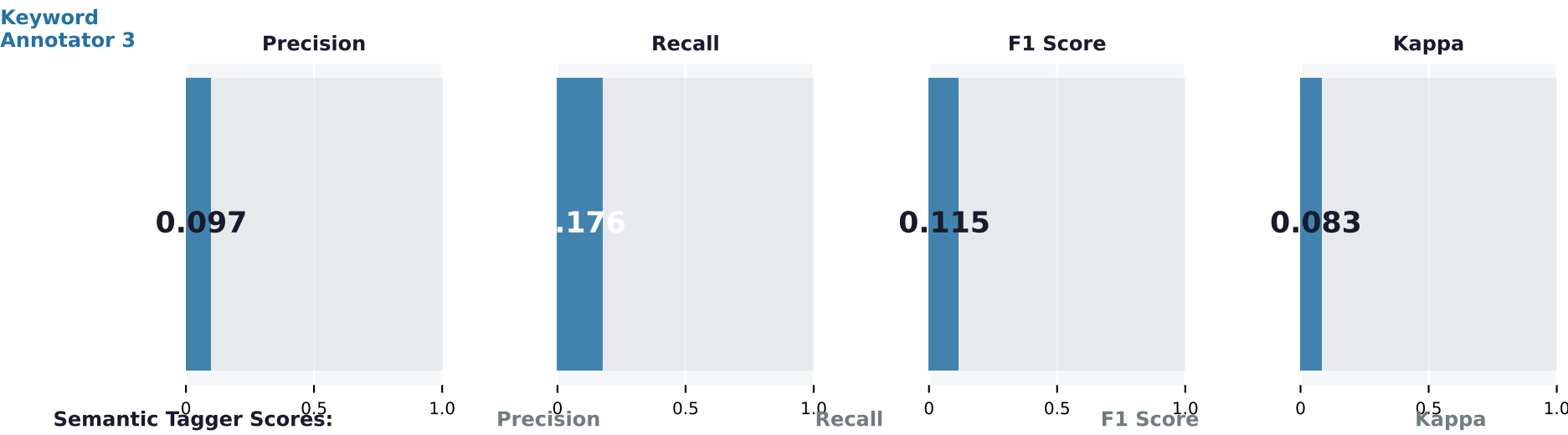
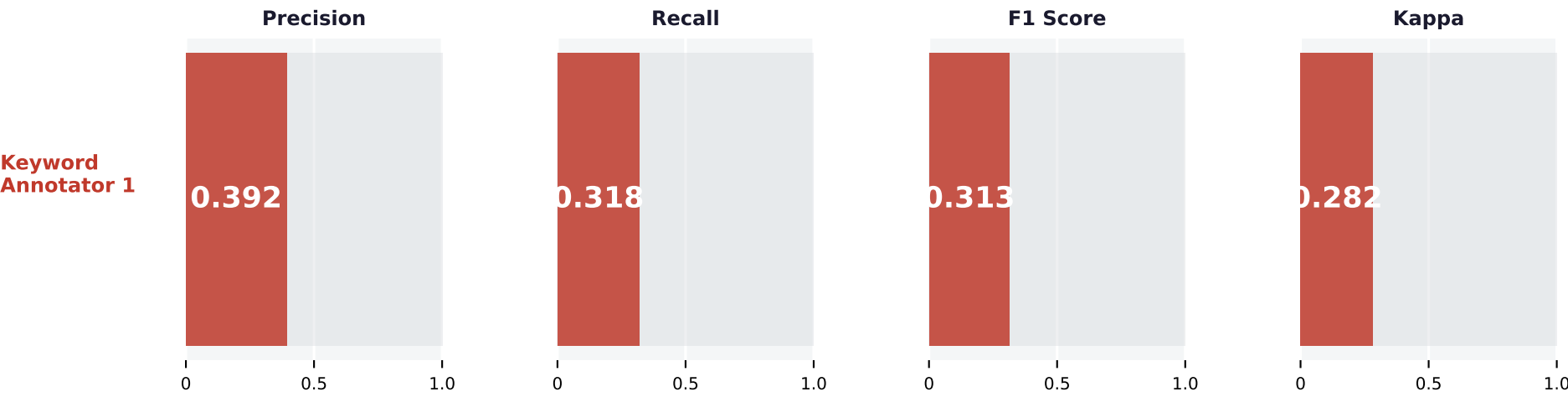


What does our result mean?

Our pipelines achieved Kappa in the 0.08-0.28 range — straddling poor to fair agreement. This is a common result for automated content analysis in communication research. It means the pipelines are doing something meaningful but are not ready to replace human coders. The recommended next step is to use them as a first pass, with human spot-checking.

Overall Performance Scorecard

Macro-averaged scores across all 12 topics for each method and annotator set.



Semantic Tagger Scores:

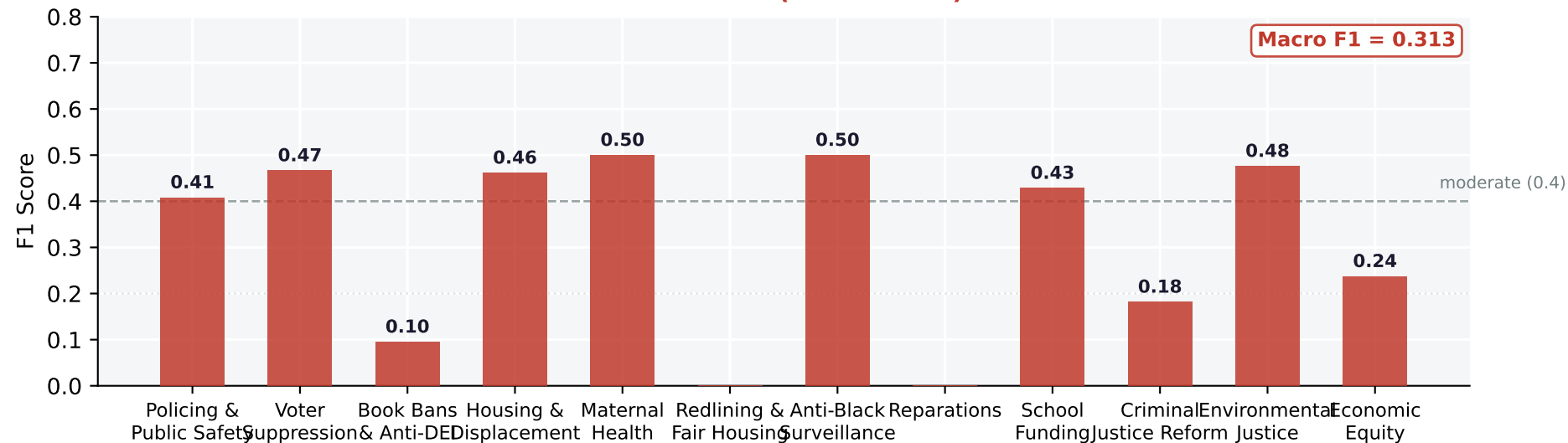
	Precision	Recall	F1 Score	Kappa
Semantic — Annotator 1	0.197	0.586	0.257	0.200
Semantic — Annotator 3	0.233	0.469	0.275	0.250

Precision: when the pipeline tags a topic, how often is it correct? Recall: of all articles a human labeled, how many did the pipeline catch?
F1: overall balance between precision and recall. Kappa: agreement corrected for chance — the most rigorous metric (≥ 0.4 = substantial, $0.2\text{--}0.4$ = fair, < 0.2 = poor).

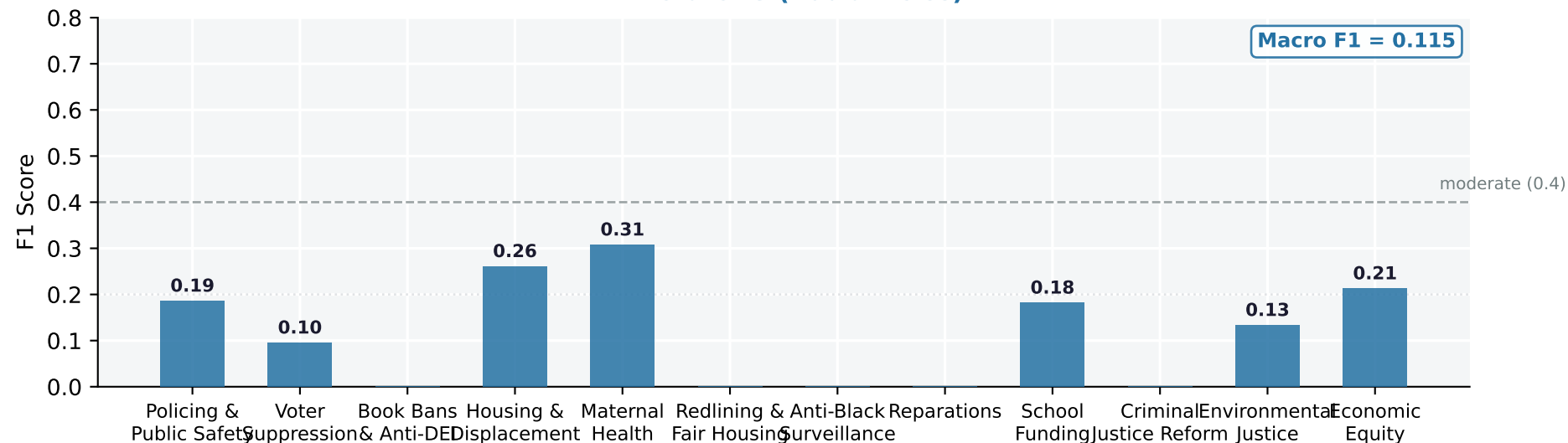
Keyword Pipeline — F1 Score by Topic

How well does the keyword pipeline agree with each annotator on each topic?

Annotator 1 (200 articles)



Annotator 3 (200 articles)

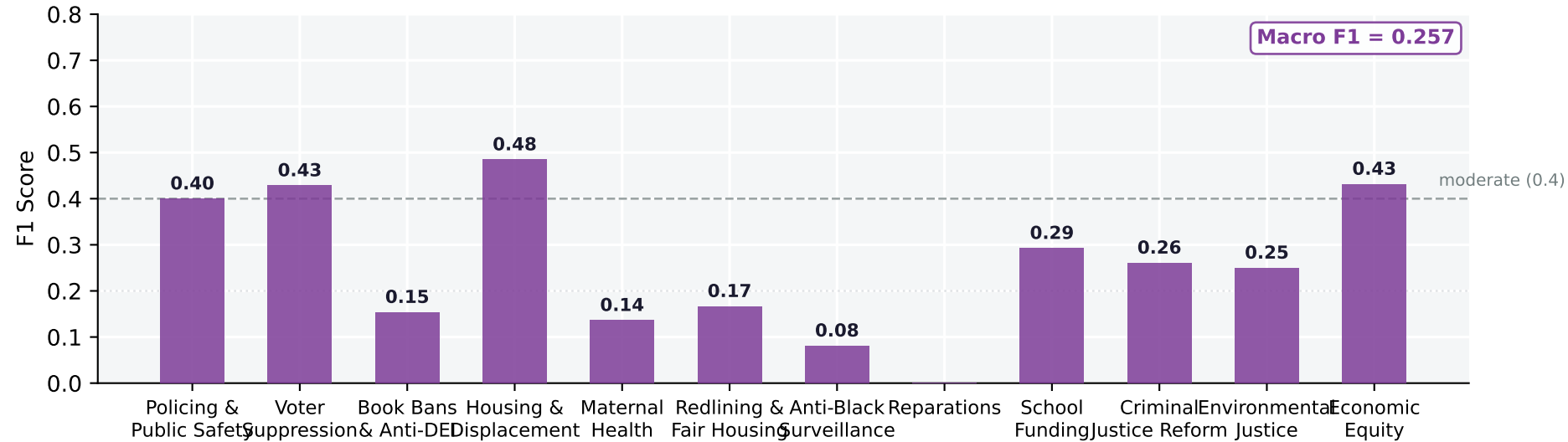


Each bar shows the F1 score for one topic — the harmonic mean of precision and recall.
A score above 0.4 (dashed line) indicates substantial agreement with the human annotator.
Topics with no bar (score = 0) were never tagged by the pipeline or never matched human labels.

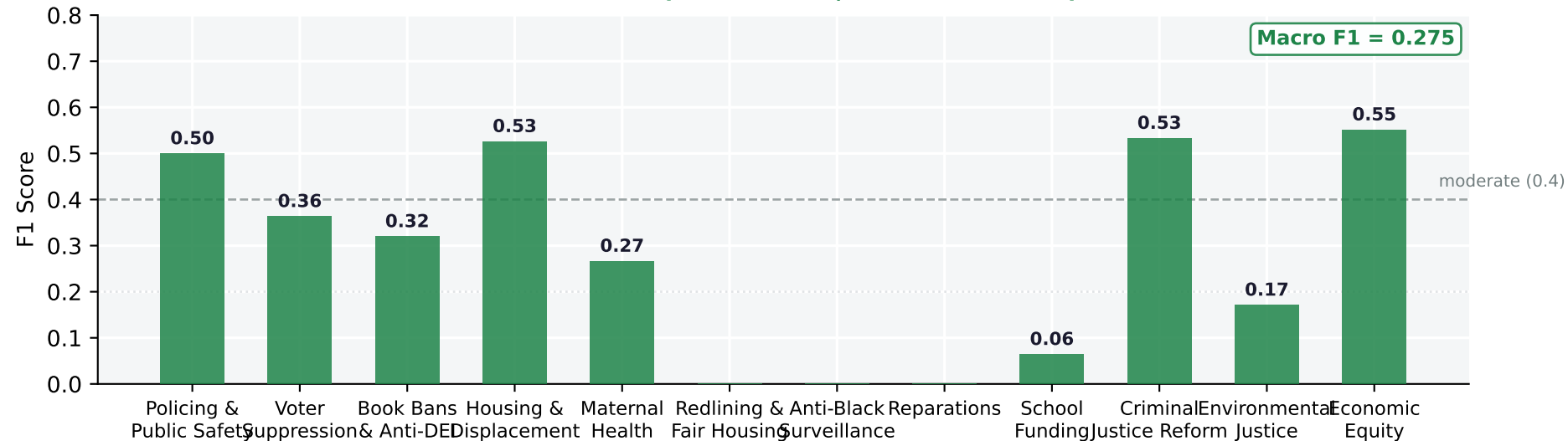
Semantic Tagger — F1 Score by Topic

How well does the semantic (AI embedding) tagger agree with each annotator on each topic?

Annotator 1 (200 articles, threshold=0.15)



Annotator 3 (200 articles, threshold=0.20)

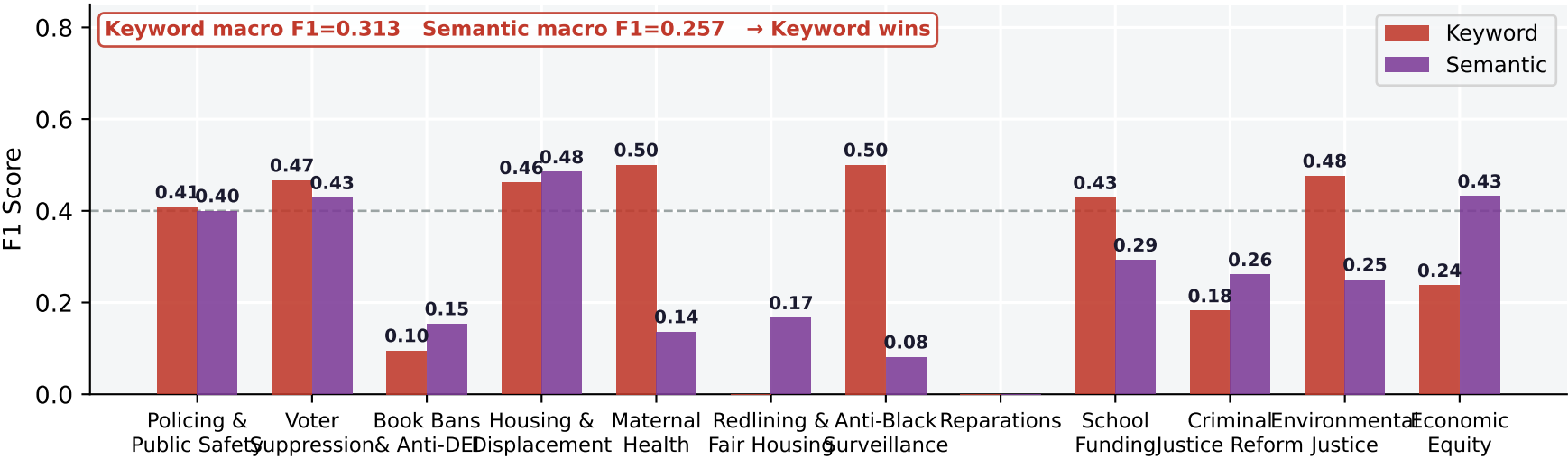


The semantic tagger converts each article and topic into an AI embedding vector, then tags articles whose similarity score exceeds the threshold (optimised per annotator). Threshold was chosen by scanning 0.10–0.35 and selecting the value that maximised macro F1.

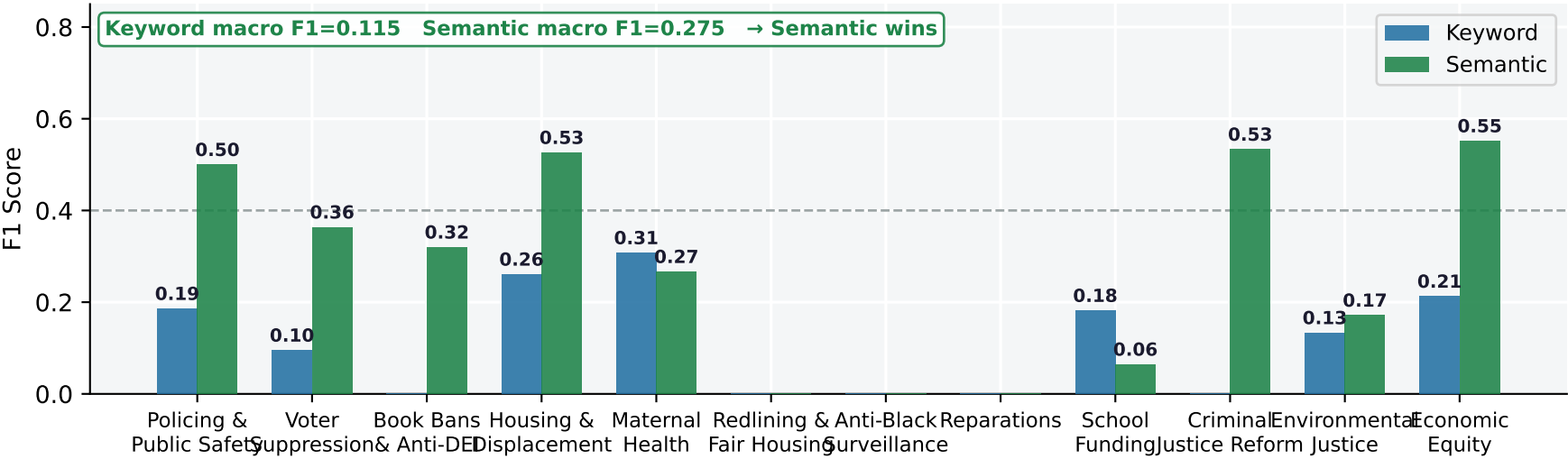
Keyword vs Semantic — Head-to-Head F1 by Topic

Direct comparison of both methods on the same article sets.

Annotator 1



Annotator 3



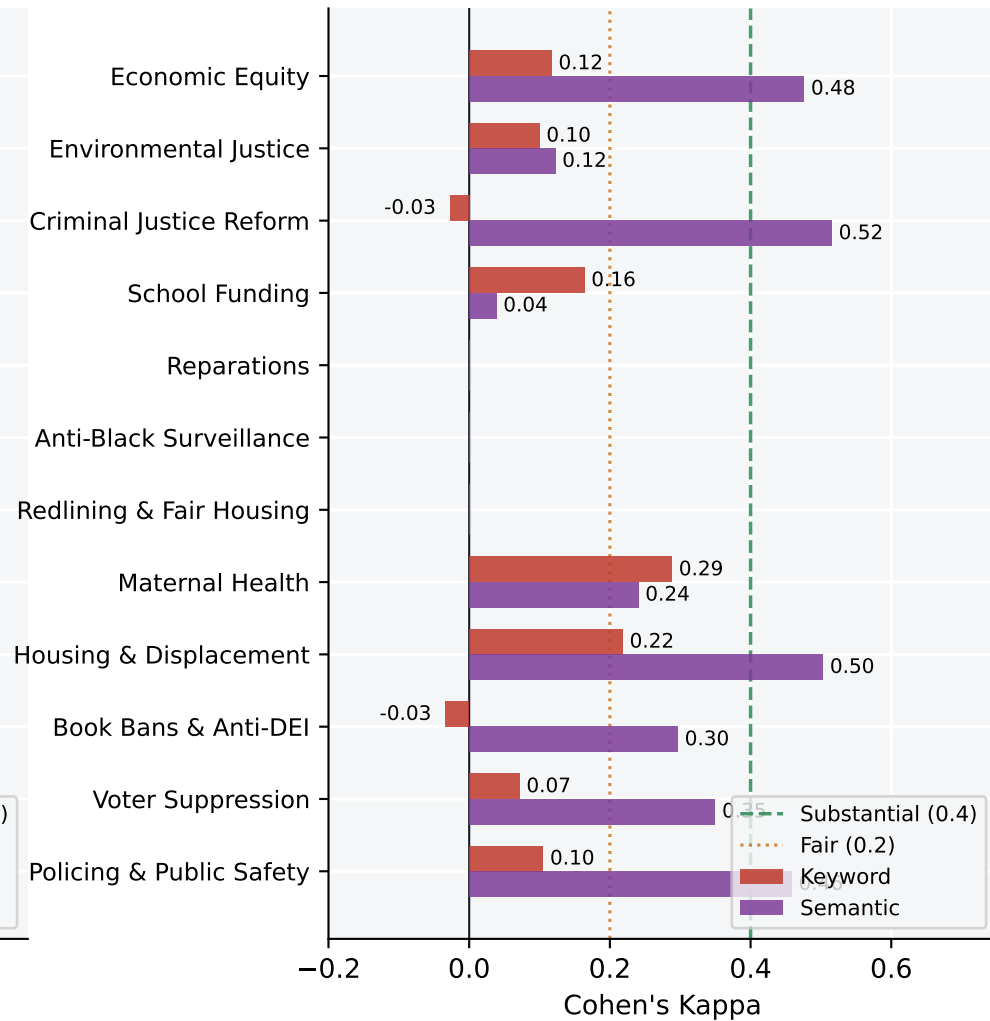
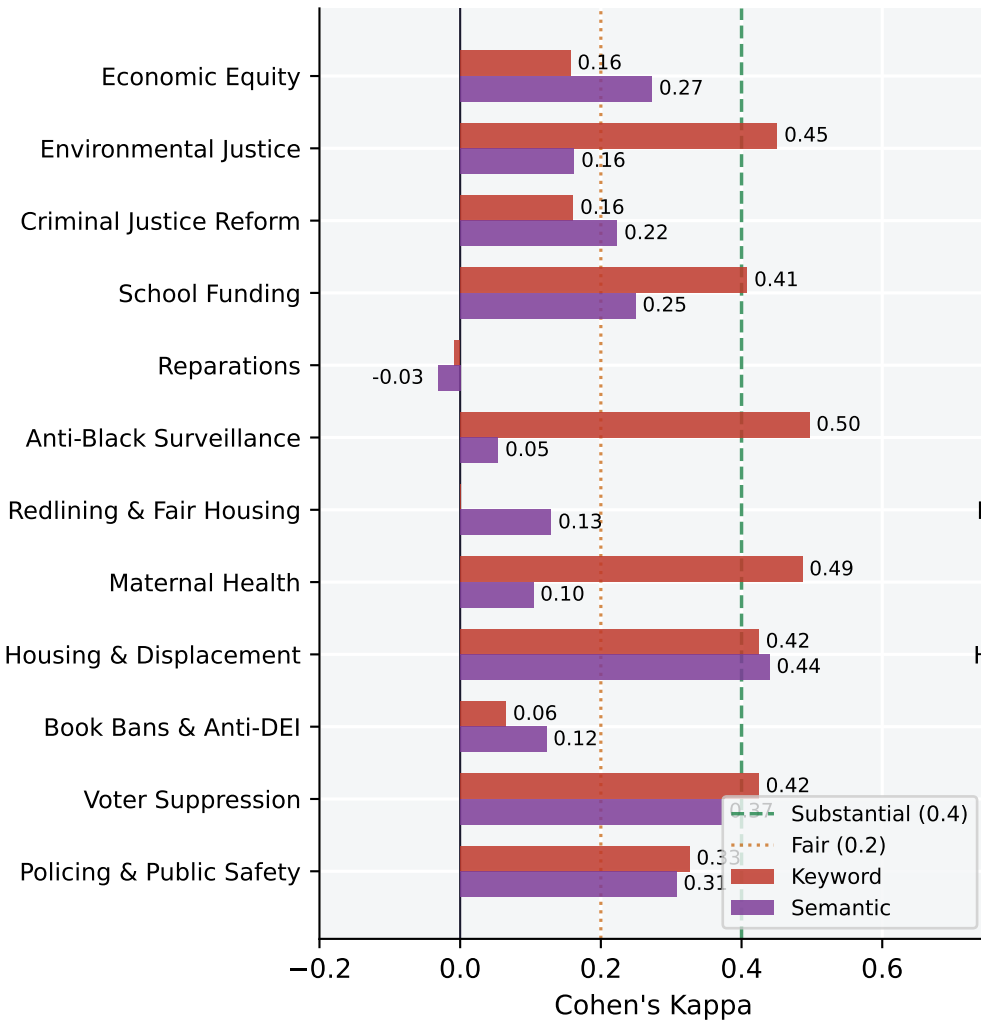
For each topic, the taller bar is the better-performing method on that annotator's article set. The keyword pipeline outperforms on Annotator 1's set; the semantic tagger wins on Annotator 3's set. This suggests neither method is universally superior — combining both may yield the best results.

Cohen's Kappa by Topic — Keyword vs Semantic

Agreement with human annotation corrected for chance. The most rigorous validation metric.

Annotator 1

Annotator 3



Substantial ($\kappa \geq 0.4$) Fair ($0.2 \leq \kappa < 0.4$) Poor ($\kappa < 0.2$)

Cohen's Kappa (κ) measures how much the pipeline and human annotator agree, beyond what would be expected by chance alone.

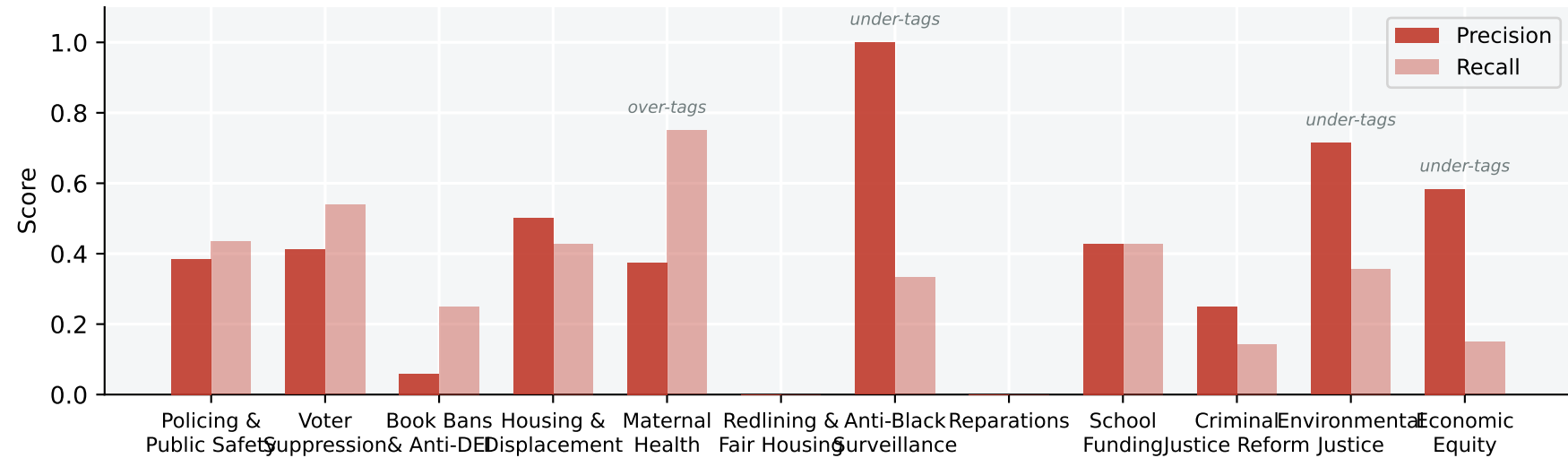
$\kappa \geq 0.4$ = substantial agreement (green dashed line) $\kappa 0.2-0.4$ = fair agreement $\kappa < 0.2$ = poor agreement.

Negative kappa means the pipeline disagrees with the human more than chance would predict.

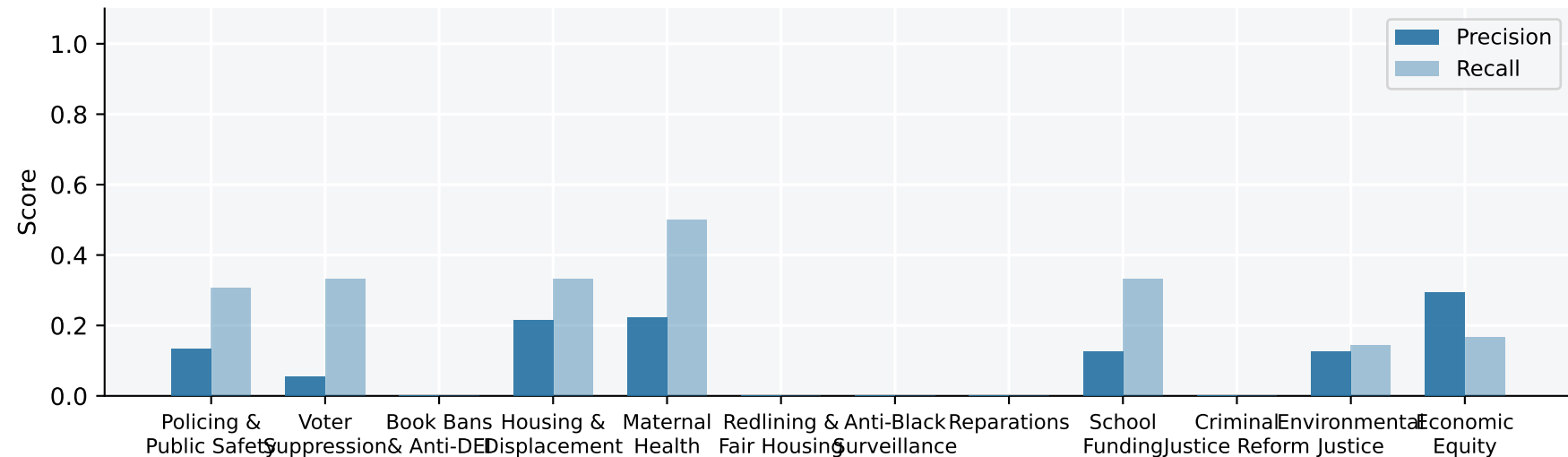
Keyword Pipeline — Precision vs Recall by Topic

Reveals whether the pipeline is over-tagging (too broad) or under-tagging (too narrow) per topic.

Annotator 1



Annotator 3



Precision (solid bar): when the pipeline assigns a topic, how often is the human in agreement?

Recall (faded bar): of all articles the human labeled for a topic, how many did the pipeline find?

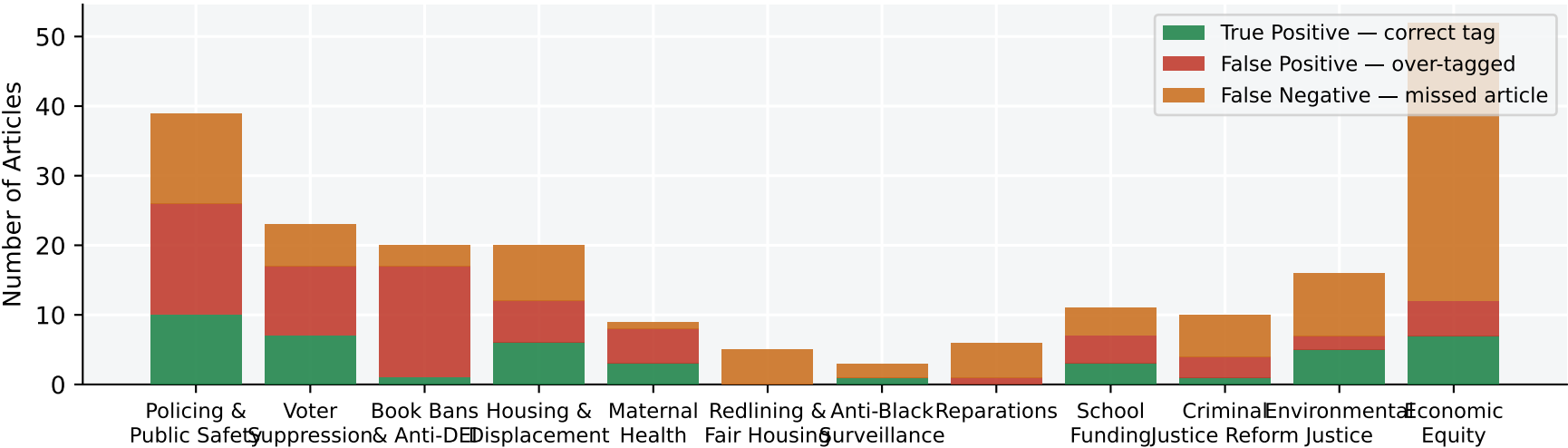
Over-tags → recall is much higher than precision (pipeline is too trigger-happy).

Under-tags → precision is much higher than recall (pipeline is too conservative, missing articles).

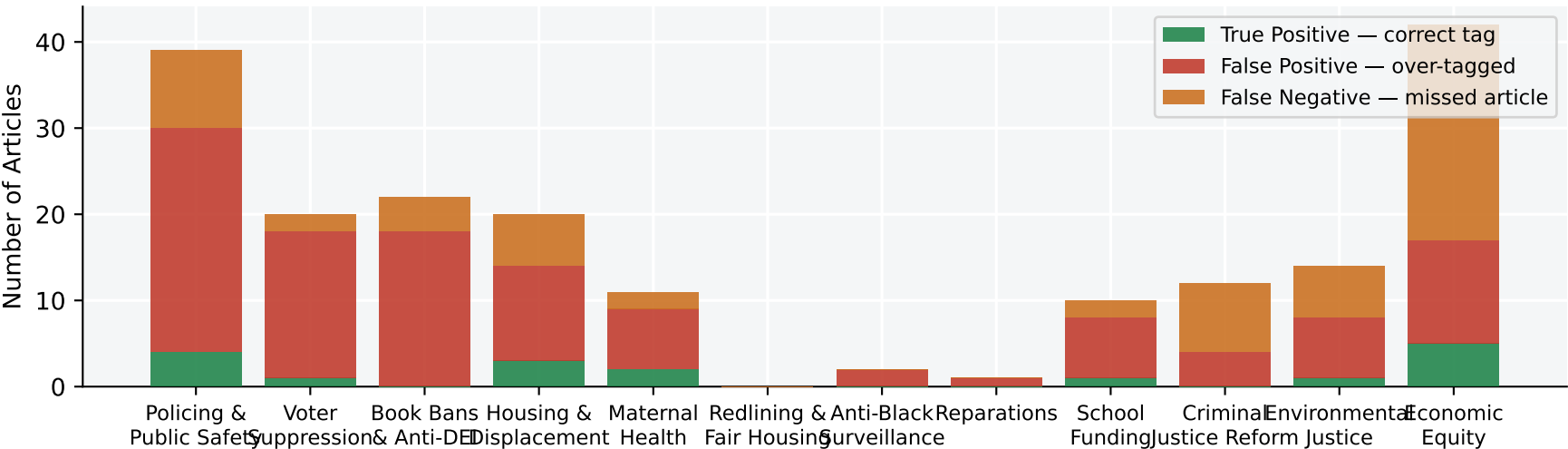
Keyword Pipeline — Error Breakdown by Topic

Exact counts of correct tags, false alarms, and missed articles per topic.

Annotator 1



Annotator 3

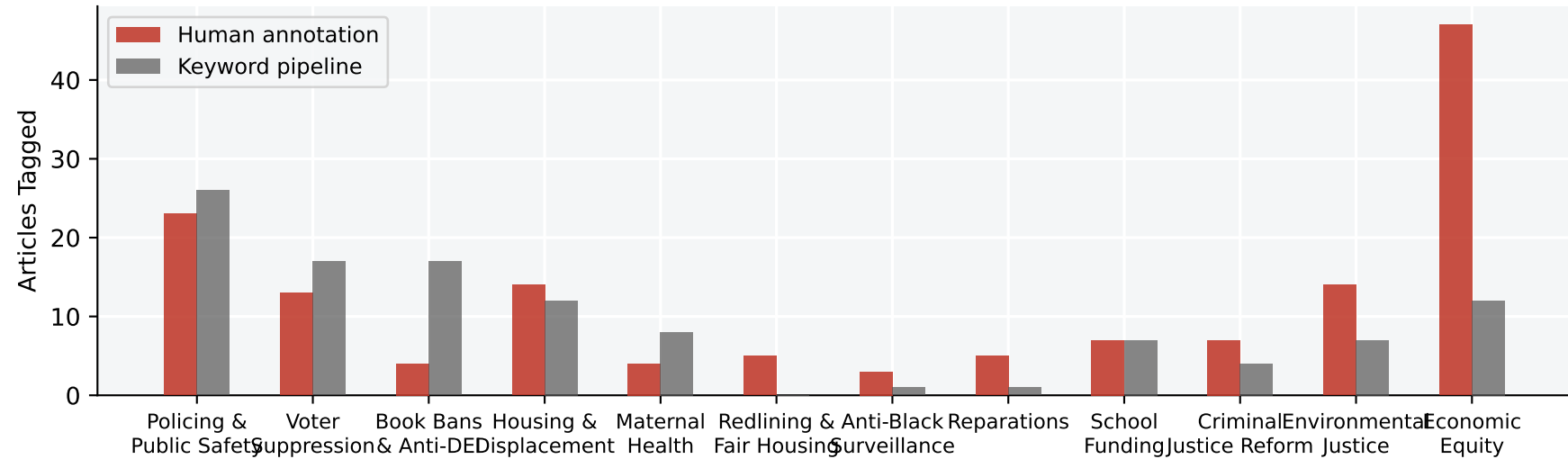


Green = both the pipeline and human agreed the topic is present (correct).
Red = pipeline tagged the topic, but the human did not (false alarm / over-tagging).
Orange = human labeled the topic, but the pipeline missed it (false negative / under-tagging).
An ideal topic has a tall green bar and minimal red and orange.

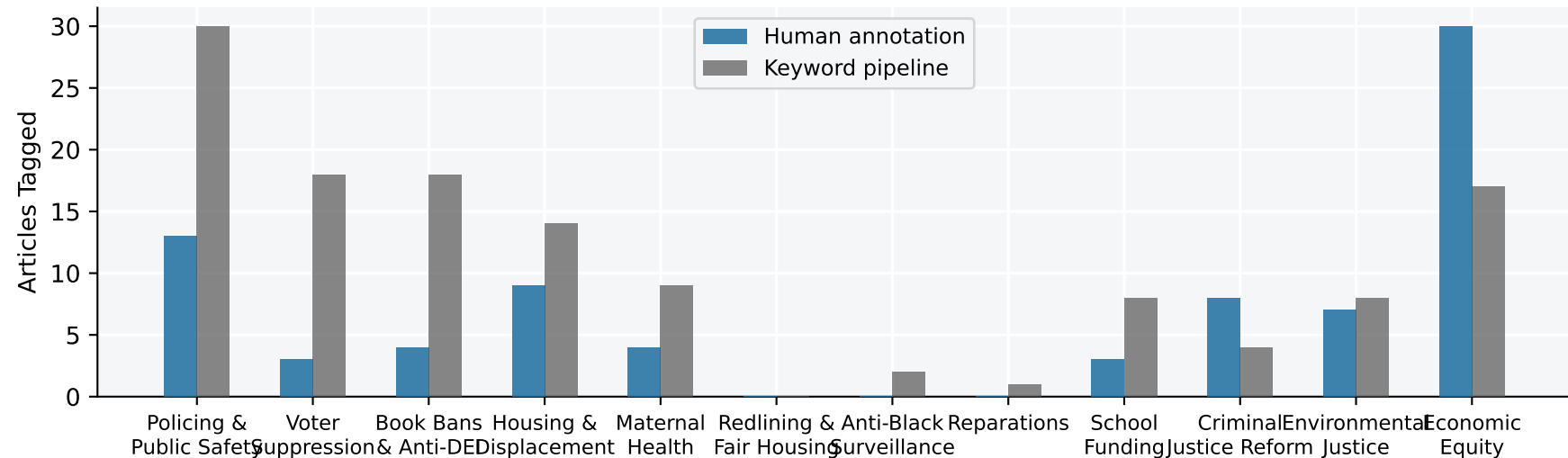
Articles Tagged per Topic — Pipeline vs Human

How many articles each method tagged for each topic vs how many the human annotator labeled.

Annotator 1 (200 articles)



Annotator 3 (200 articles)



This chart shows volume — how many articles were tagged per topic, not whether they were the right articles.

If the pipeline bar is much taller than the human bar, the pipeline is over-tagging.

If the pipeline bar is much shorter, the pipeline is under-tagging.

Equal heights do not guarantee accuracy — the pipeline may be tagging different articles than the human.

Introducing the Hybrid Tagger

A third approach that combines the strengths of both baselines.

Hybrid OR (Union)

Tags an article if the keyword pipeline
OR the semantic tagger fires.

Effect: maximum recall — nothing gets
missed that either method would catch.

Trade-off: precision falls because both
methods' false positives accumulate.

Hybrid AND (Intersection)

Tags an article only if BOTH the keyword
pipeline AND the semantic tagger agree.

Effect: maximum precision — only tags
the clearest, most confident cases.

Trade-off: recall drops sharply; many
real articles are missed.

Hybrid Soft Boost ↗ Primary

Keyword matches are always kept.
Semantic is only allowed to add a tag
when its score $\geq$ a per-topic threshold
optimised by grid search.

Effect: keyword anchors precision;
semantic fills in recall gaps — but
only when it is sufficiently confident.

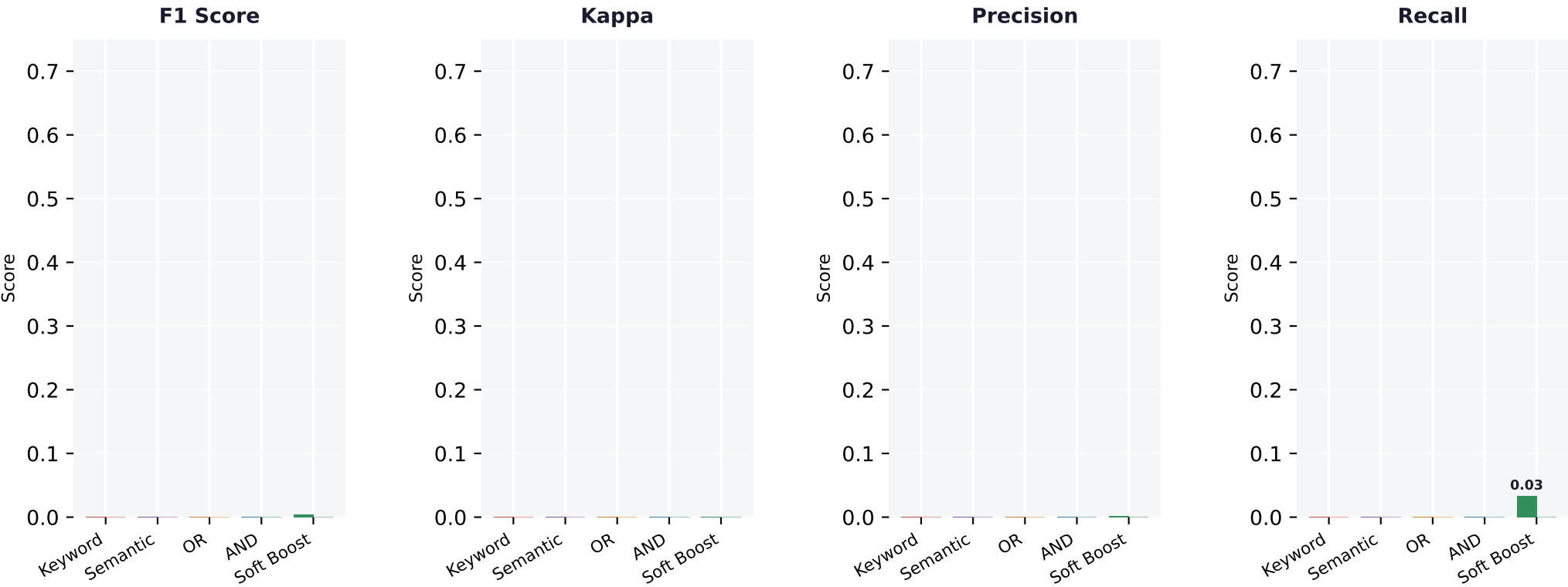
Thresholds range from 0.05 to 0.38
depending on how well-calibrated the
semantic model is per topic.

Design rationale

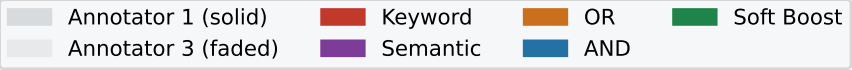
The keyword pipeline is precise but misses articles that use different wording. The semantic tagger has strong recall but over-tags. The Soft Boost hybrid is designed to keep keyword's reliable hits while using the semantic score as a calibrated second opinion, activated only when its confidence exceeds a per-topic threshold learned from the annotation data.

All Five Methods — Macro-Averaged Scorecard

Macro-averaged across all 12 topics. Each bar pair shows Annotator 1 (left) and Annotator 3 (right).



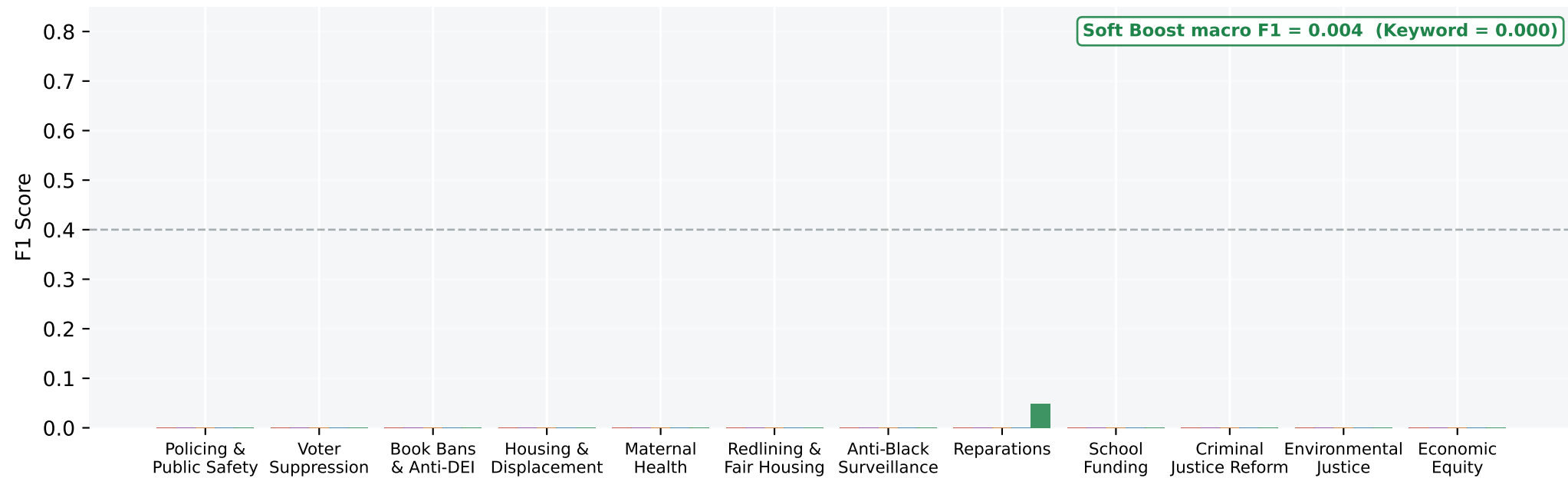
Soft Boost consistently outperforms all other methods on both F1 and Kappa.
OR (Union) maximises recall but drops precision. AND (Intersection) maximises precision but loses recall.
Soft Boost finds the best balance by keeping keyword anchors and using semantic as a calibrated second opinion.



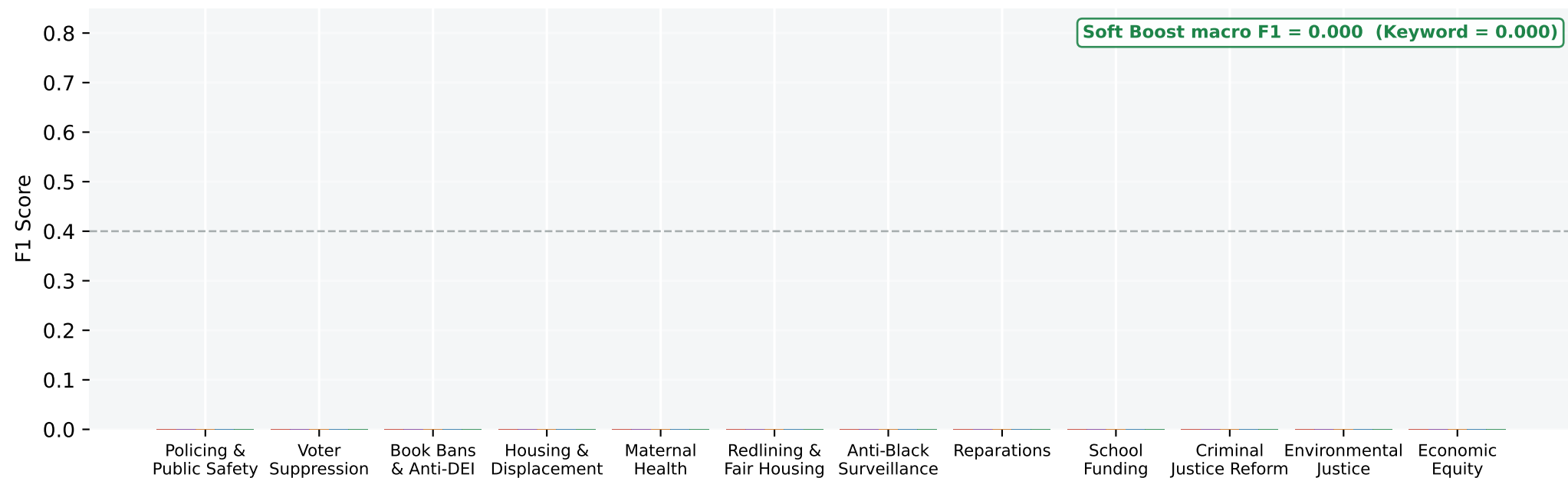
Hybrid Strategies — F1 Score by Topic

Comparing all three hybrid variants against the two baselines, per topic.

Annotator 1



Annotator 3

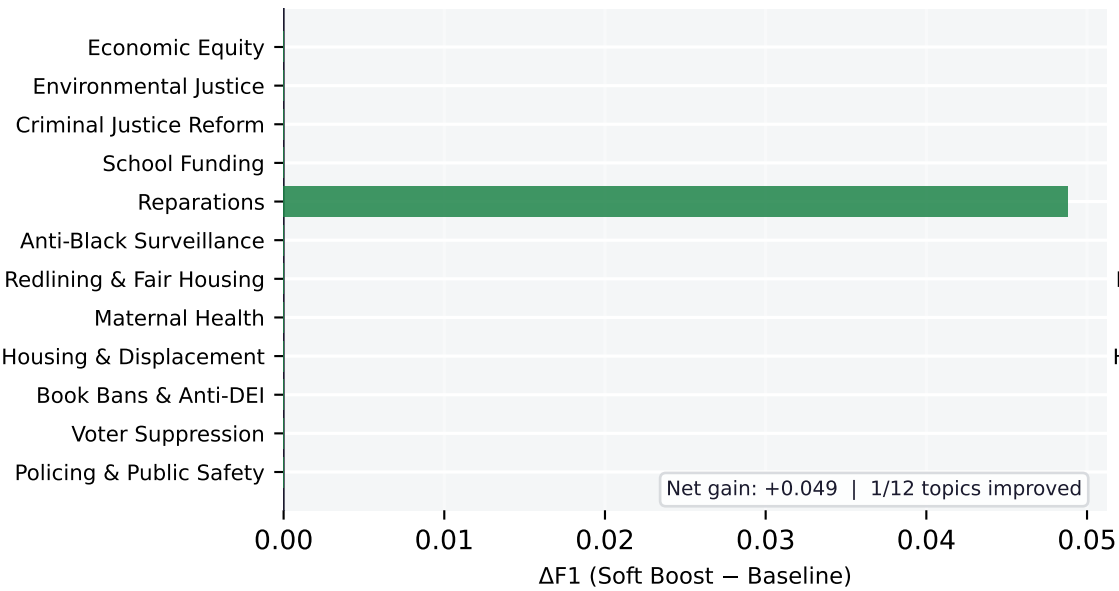


Keyword (baseline) Semantic (baseline) Hybrid OR Hybrid AND Hybrid Soft Boost ♦

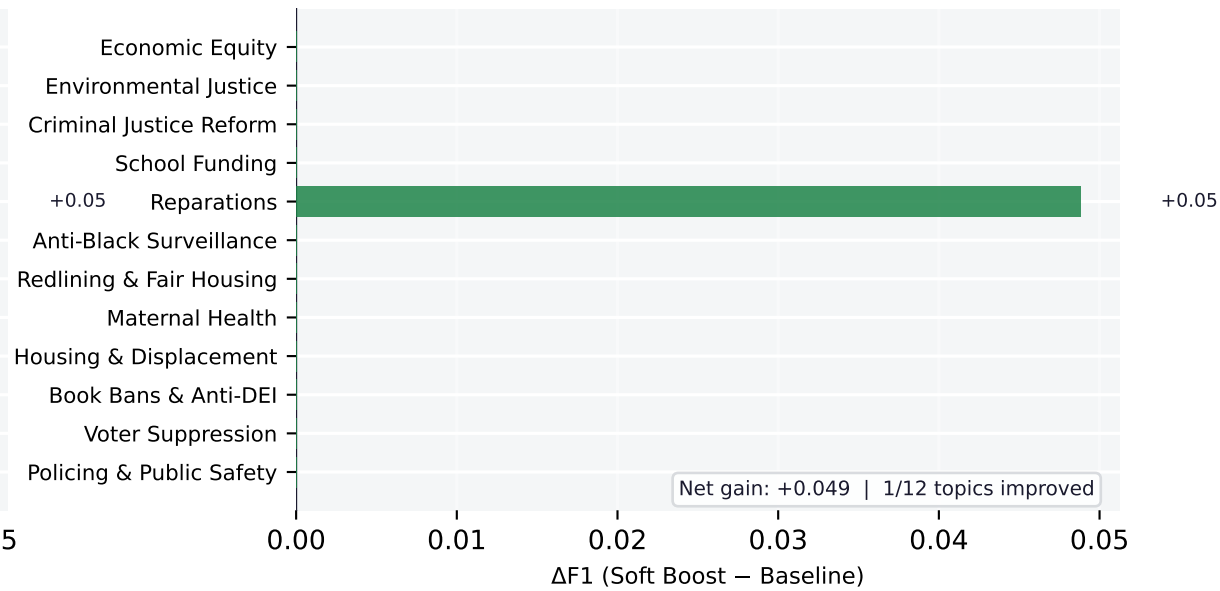
Hybrid Soft Boost — Gain Over Baselines ($\Delta F1$)

Per-topic F1 improvement of Soft Boost over keyword (left) and semantic (right). Green = improvement, red = regression.

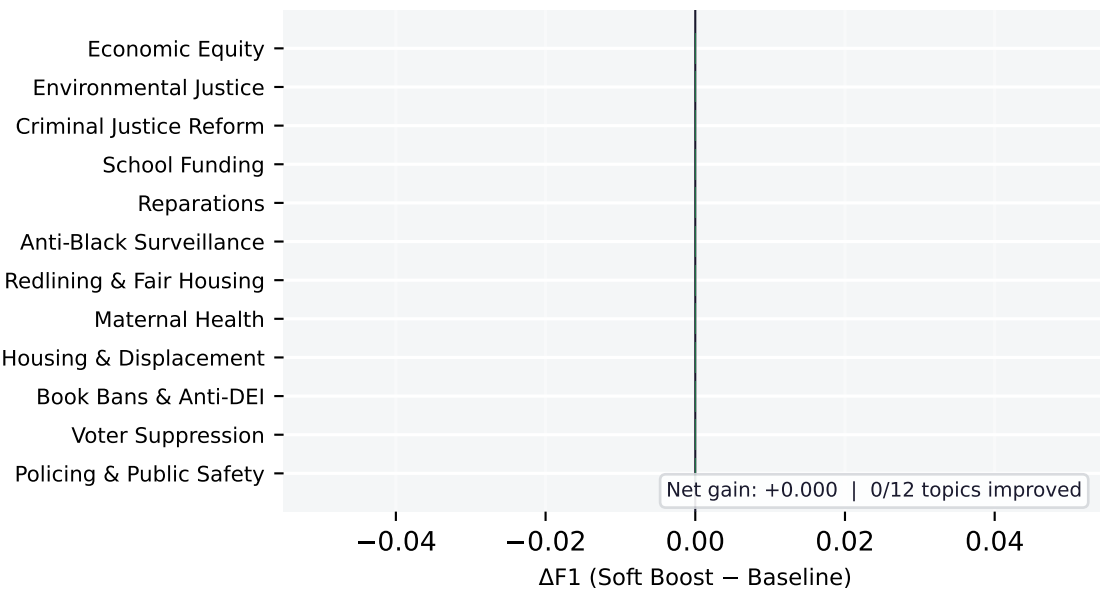
Annotator 1 — vs Keyword



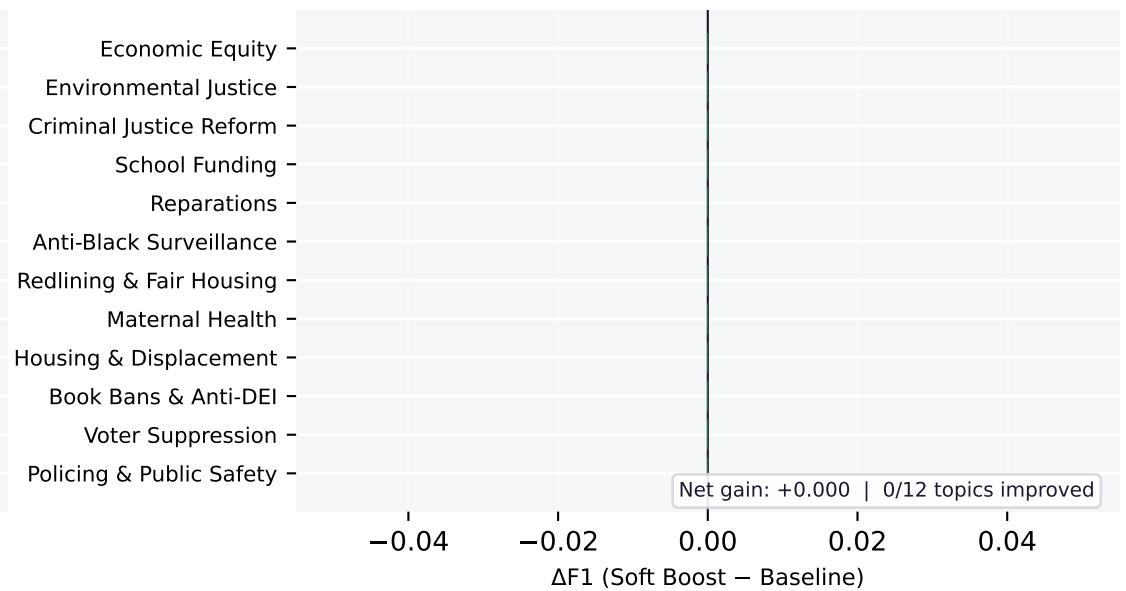
Annotator 1 — vs Semantic



Annotator 3 — vs Keyword



Annotator 3 — vs Semantic



A positive bar means Soft Boost outperforms the baseline on that topic. Negative means the hybrid is worse (e.g. where keyword alone is already optimal).
Topics with zero delta had no articles tagged by either method.